

Software Development

Programming Tools

IDENTIFICATION

CODE : IF-3-OP
ECTS : 0.0

HOURS

Lectures : 4.5 h
Seminars : 4.0 h
Laboratory : 4.0 h
Project : 0.0 h
Teacher-student
contact : 12.5 h
Personal work : 10.0 h
Total : 22.5 h

ASSESSMENT METHOD

None

TEACHING AIDS

Lecture notes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course introduces the Linux file systems (on the user side - no administration). It also shows how to build a program using the basic tools (make, gdb, valgrind).

Skills

Target skills are as follow:

- Being able to use an OS and its basic commands for programming;
- Mastering the compilation chain of a program;
- Using generic tools to improve software quality.

CONTENT

This course is divided into 2 main chapters:

1. Quick introduction to the Linux operating system (on the user side):

- Short history;
- The Linux file system (disk organization, types of files, i-nodes, access rights, basic commands);
- The working session (list of important files).

2. Program building and main tools:

- Compilation (compilation steps, compiler options);
- Interpretation;
- make command make and the makefile file (syntax and use);
- gdb command (main commands to debug a program);
- valgrind command (memory management monitoring).

BIBLIOGRAPHY

- [1] Tanenbaum Andrew. Modern operatings systems. Upper Saddle River: Pearson International
- [2] Silbershatz Abraham. Operating system concepts. USA: John Wiley & Sons inc
- [3] Bourne-Again SHell manual, Bash Reference Manual, Free Software Foundation, <http://www.gnu.org/software/bash/manual>
- [4] GNU Make manual, Make Reference Manual, Free Software Foundation, <http://www.gnu.org/software/make/manual>
- [5] GDB: The GNU Project Debugger, Free Software Foundation, <http://www.gnu.org/software/gdb/documentation>
- [6] Valgrind User Manual [release 3.13.0 - 15 juin 2017], Valgrind Developers, <http://valgrind.org/docs/manual/manual.html>

PRE-REQUISITE

None

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